



Department of Computer Science and Engineering

AI-Agent–Integrated Framework for Optimizing Administrative Workflows Across Clinical and Health Insurance Systems

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(IV -CSE A1)

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Aim & Objective

Aim

- To develop an AI-agent-integrated framework that optimizes and automates administrative workflows across clinical and health insurance systems

Objective

- To integrate clinical and insurance workflows into a unified intelligent system.
- To use multiple AI agents for symptom analysis, diagnostic recommendation, treatment evaluation, claim decision, and explanation generation.
- To improve administrative efficiency by reducing manual intervention and repetitive tasks.
- To provide real-time, transparent, and explainable decision-making.

Abstract

An **AI-Agent-Integrated Framework** is proposed to streamline fragmented healthcare and insurance processes. It uses multiple specialized AI agents—Symptom Interpretation, Diagnostic Recommendation, Treatment Evaluation, Claim Decision, and Explanation—to work together. These agents combine large language models with rule-based checks to analyze symptoms, suggest tests, assess treatments, process insurance claims, and provide clear explanations, reducing delays, inefficiencies, and operational costs.

Introduction

- Healthcare has improved clinically, but administrative workflows are still fragmented.
- Key processes (symptom assessment, diagnosis, treatment validation, insurance claims) are handled separately.
- This leads to duplication, delays, higher costs, and low transparency.

AI advancements (multi-agent systems, generative AI) can improve efficiency.

- Proposed solution: a modular multi-agent framework connecting clinical care with insurance processes.

Literature Review

S.NO	Authors	Paper Title	Method Used	Contribution
1	V. G. Hinostroza Fuentes et al.	AI with Agency: A Vision for Adaptive, Efficient, and Ethical Healthcare	Multi-agent AI systems, Generative AI	Proposes adaptive and ethical AI agents for integrated healthcare workflows
2	M. Usman et al	An Intelligent AI base Framework for healthcare Application	AI framework, Machine Learning models	Presents a general AI-based architecture for improving healthcare applications
3	P. Niraula et al.	AI and Machine Learning in Digital Health Systems	Machine Learning, Data-driven models	Reviews ML applications in digital health systems
4	X.Zhou at al.	AI-Based Data Analysis in Healthcare and Environmental Health Study	Data analytics, AI models	Demonstrates AI for healthcare and environmental data analysis

System Analysis

Existing System

- Clinical and insurance systems work in isolation
- Administrative processes rely heavily on manual work
- Limited use of real-time intelligence
- Lack of transparency in decision-making

Disadvantages

- No proper integration between hospital and insurance workflows
- High manual effort in processes
- Increased errors and inconsistencies
- Delays in diagnosis and claim approvals
- Repetitive data entry tasks

System Analysis

Features

- Multi-agent AI architecture
- Symptom analysis and severity detection
- Diagnostic recommendation
- Treatment validation

Advantages

- Seamless workflow integration
- Reduced manual effort
- Faster decision-making
- Better accuracy and consistency

System Requirements

Hardware Requirement

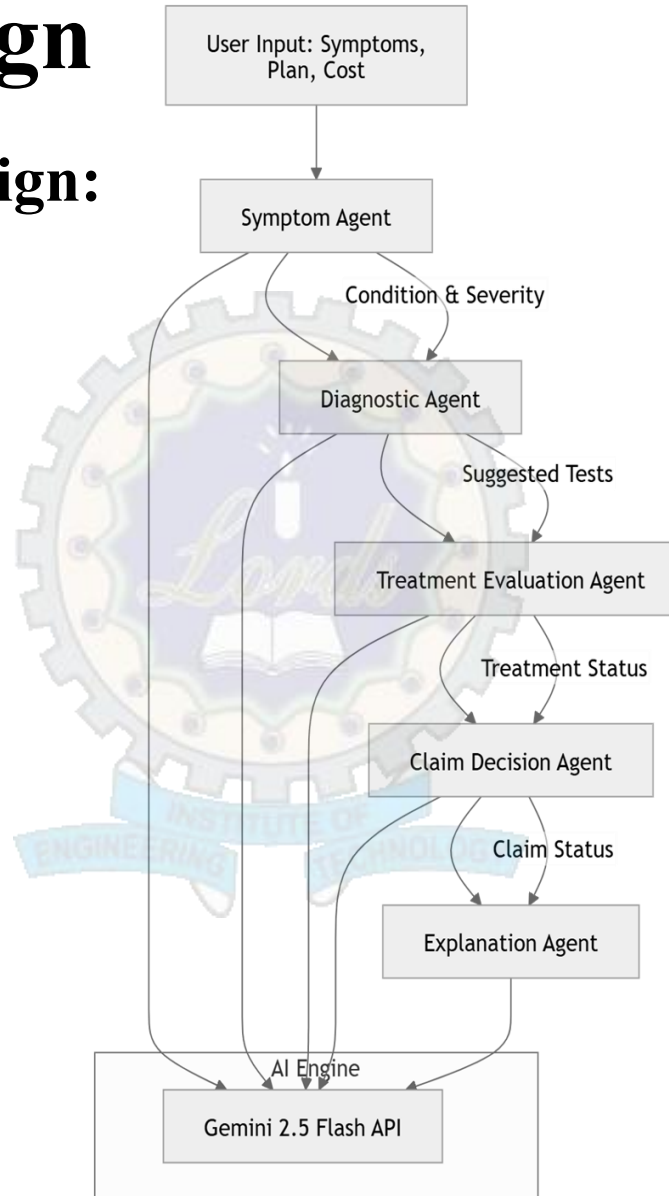
- System: Pentium i3 Processor
- Hard Disk: 500 GB.
- Monitor: 15'' LED
- Input Devices: Keyboard, Mouse
- Ram: 2 GB

Software Requirement

- Operating System: Windows 10.
- Coding Language: Python 3.10
- Web Framework: Flask

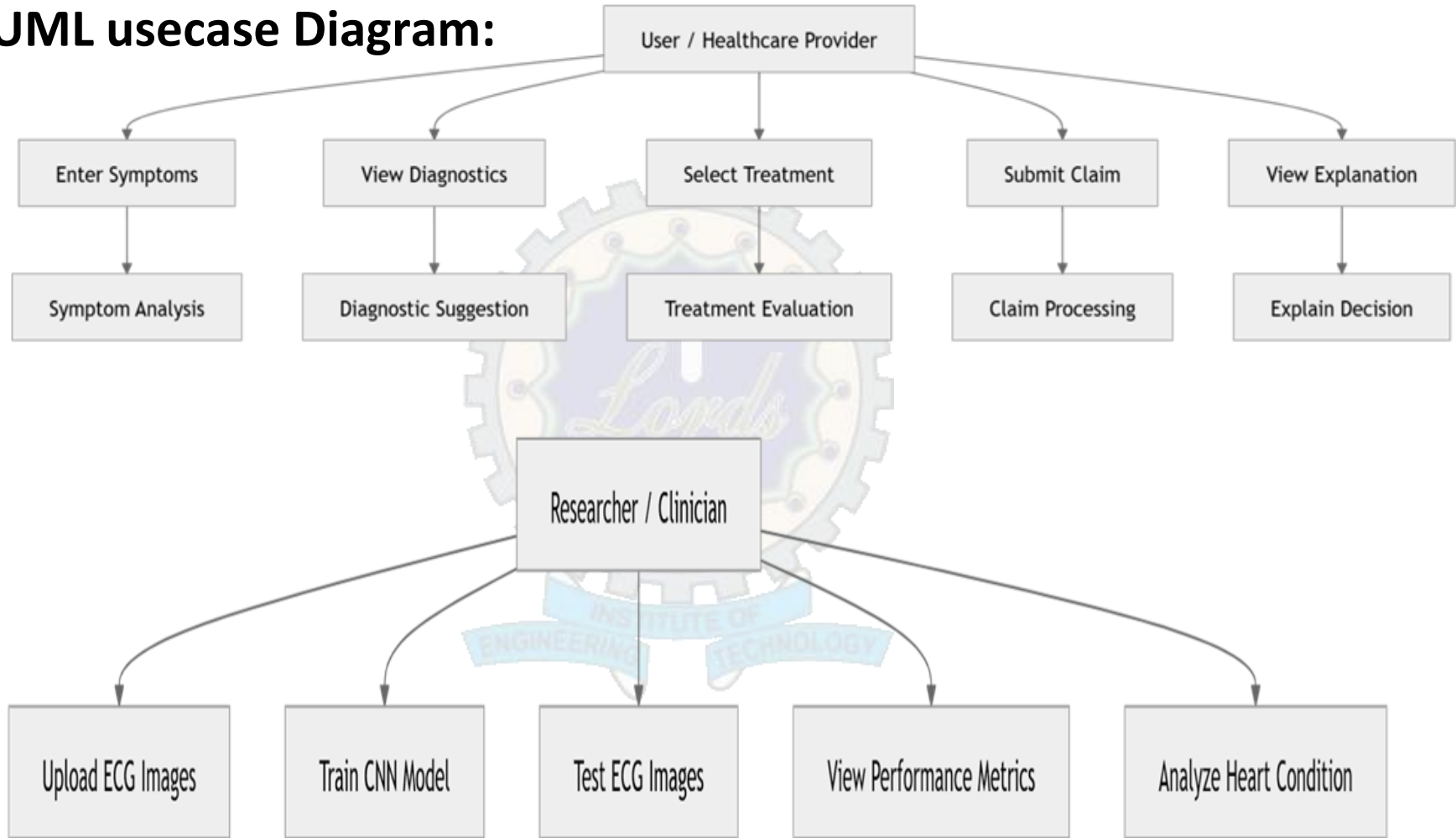
System Design

Architecture Design:



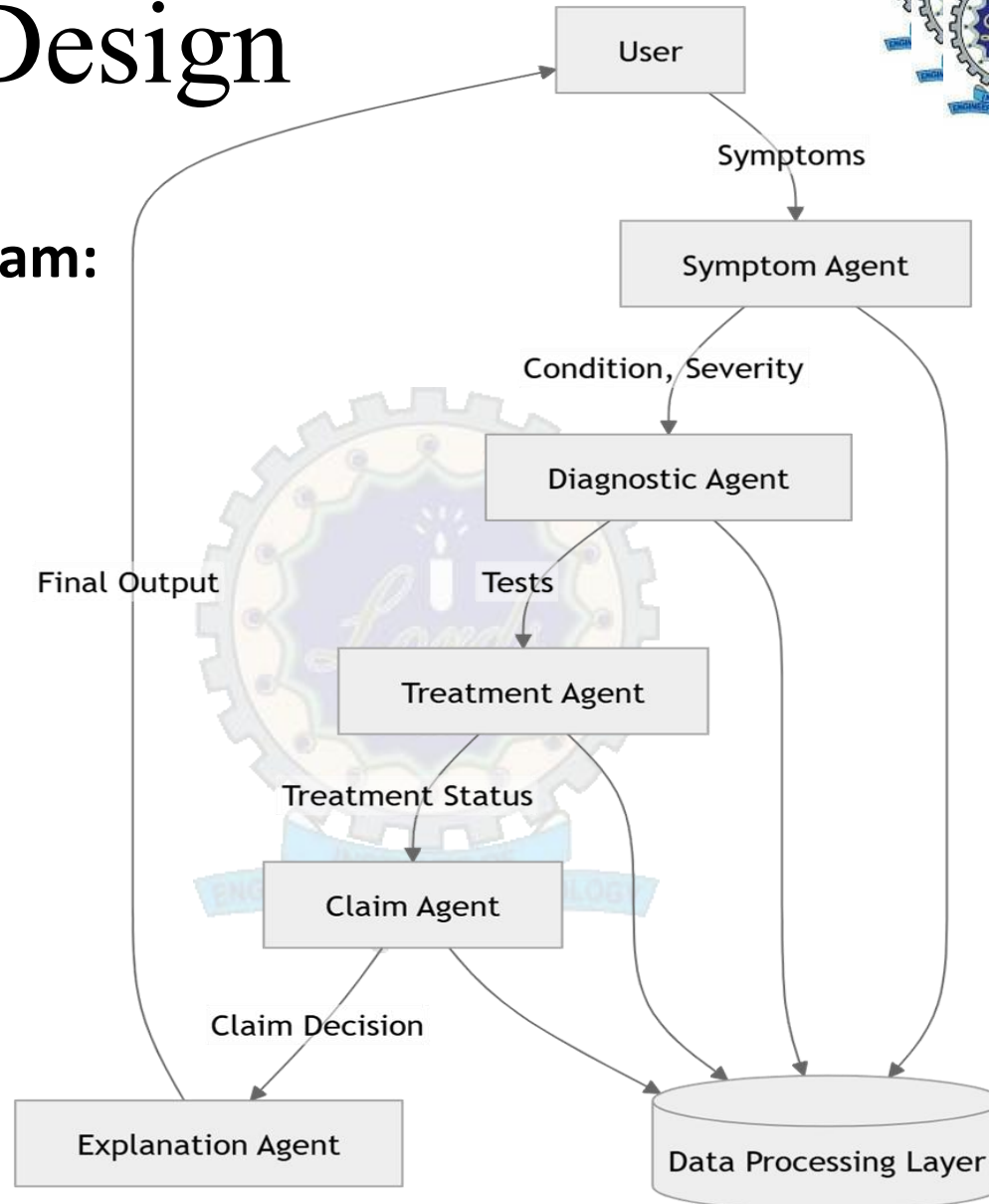
System Design

UML usecase Diagram:



System Design

Data flow Diagram:



System Implementation

Methodology

The system works in the following steps:

- Collect patient symptoms, treatment type, plan, and cost
- Analyze symptoms using the Symptom Agent
- Recommend tests through the Diagnostic Agent
- Evaluate treatment appropriateness using the Treatment Agent
- Decide claim approval using the Claim Agent
- Generate human-readable summary using the Explanation Agent
- Display complete results to the user through the interface

System Implementation

Technology used

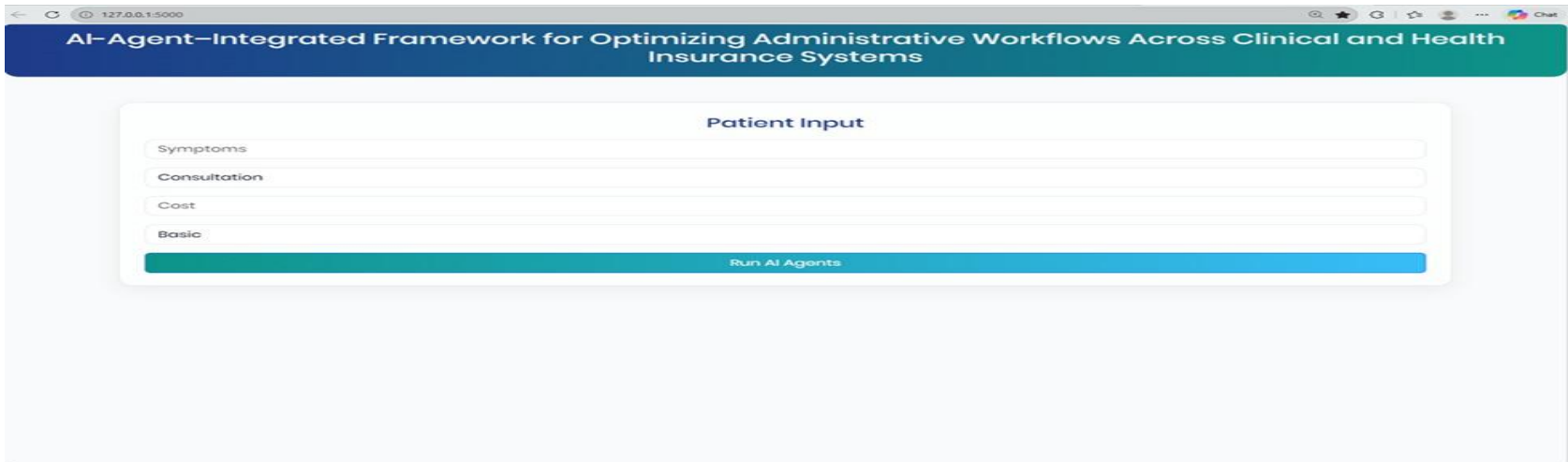
- Python 3.10
- Flask
- JSON
- Regular Expressions
- Gemini 2.5 Flash API

Algorithms used

- Multi-Agent System (MAS)
- Large Language Models (LLM)
- Natural Language Processing (NLP)
- Rule-Based Algorithm

Results

- Screenshots



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Patient Input

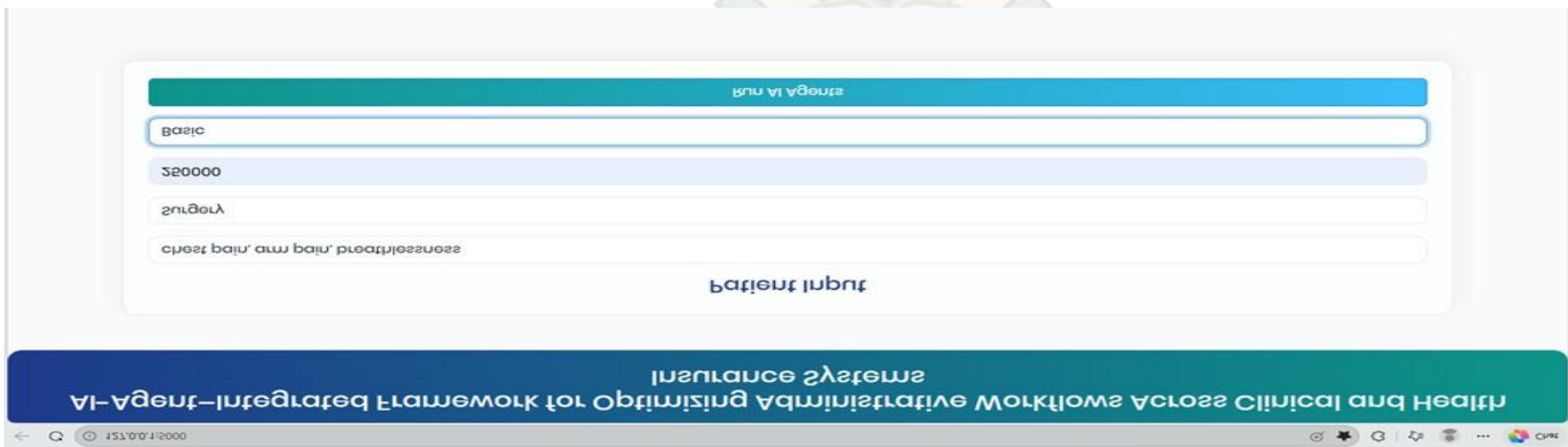
Symptoms

Consultation

Cost

Basic

Run AI Agents



Run AI Agents

Basic

320000

Surgery

chest pain, arm pain, breathlessness

Patient Input

Insurance Systems

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Results

- Screenshots

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AI Agent Processing

Symptom Agent

Processing...

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AI-Agent-Integrated Framework for Optimizing Administrative Workflows Across Clinical and Health Insurance Systems

AI Agent Processing

Symptom Agent

Condition: Myocardial Infarction (Heart Attack)

Severity: **Critical**

Confidence: 95%

Diagnostic Agent

Recommended Tests

- Electrocardiogram (ECG)
- Cardiac Troponin

Confidence: 98%

Treatment Agent

Evaluation: **Appropriate**

Reason: Surgical intervention is often necessary to restore blood flow and salvage heart muscle during a Myocardial Infarction.

Confidence: 90%

Claim Agent

Claim Status: **Partial Approval**

Reason: Cost exceeds basic plan

Confidence: 85%

Explanation Agent

Summary:

- Condition identified as Myocardial Infarction (Heart Attack) with Critical severity.
- Electrocardiogram (ECG) and Cardiac Troponin tests are suggested.
- Surgical intervention is appropriate to restore blood flow and salvage heart muscle.
- Claim is partially approved as cost exceeds the basic plan.

Confidence: 92%

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THANK YOU

